

About the Author



Banu Parasuraman is a seasoned technologist with over 30 years of experience in the IT industry, currently focusing on the transformative fields of AI and automation. Throughout his distinguished career, Banu has provided invaluable advisory services to clients across a multitude of industries, guiding them in leveraging AI-driven solutions and automation technologies to enhance their operations and achieve digital transformation. His expertise has been sought by over 25 leading companies spanning sectors like retail, healthcare, logistics, banking, manufacturing, automotive, oil and gas, pharmaceuticals, media, and entertainment across the United States, Europe, and Asia.

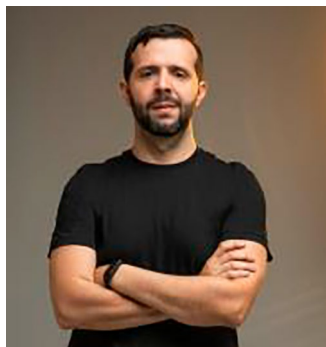
A passionate advocate for AI and automation, Banu has been at the forefront of encouraging organizations to adopt these technologies early, helping them to streamline operations, improve decision-making, and maintain a competitive edge. His deep knowledge extends across all major AI and automation platforms, including advanced cloud services and tools that integrate seamlessly into enterprise environments.

Banu is a respected speaker, having participated in numerous external engagements such as VMworld, SpringOne, Spring Days, and Spring Developer Forum Meetups, where he shared his insights with CXOs and engineers alike. His internal engagements include leading workshops on AI-driven solutions and automation strategies, where he has been instrumental in enabling sales plays and strategies focused on these cutting-edge technologies.

ABOUT THE AUTHOR

Beyond speaking engagements, Banu is an active thought leader in the industry, contributing to the wider adoption of AI and automation through his blogs on Medium and LinkedIn, where he shares his knowledge and promotes best practices in AI and automation development. He is also the author of the book *Practical Spring Cloud Function: Developing Cloud-Native Functions for Multi-Cloud and Hybrid-Cloud Environments*, available on Amazon. Banu's work continues to shape the future of technology, inspiring organizations to innovate and embrace the power of AI and automation.

About the Technical Reviewers

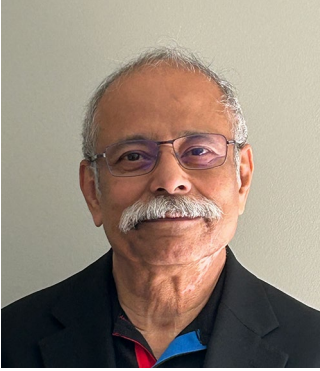


Andres Sacco has been a professional developer since 2007, working with various languages, including Java, Scala, PHP, Node.js, and Kotlin. His background is mainly in Java and its associated libraries or frameworks, like Spring, JSF, iBATIS, Hibernate, and Spring Data. He is focused on researching new technologies to improve the performance, stability, and quality of the applications he develops.

In 2017, he started to find new ways to optimize data transference between applications to reduce the cost of infrastructure. He suggested some actions, some applicable to all the microservices and others to just a few of them; as a result, the cost was reduced by 55%. Some of these actions are connected directly with the bad use of the databases.

He recently published some books with Apress about the last version of Scala and Spring Data. He also published a set of theoretical-practical projects on uncommon testing methods, such as architecture tests and chaos engineering.

ABOUT THE TECHNICAL REVIEWERS



Debasish Banerjee, Ph.D., is a seasoned thought leader, hands-on architect, and practitioner of cutting-edge technologies. He has a proven track record spanning more than a decade of advising and closely working with Fortune 500 and other customers in the United States, Europe, and Asia. He recently joined Guild Systems Inc (www.guildsystems.com) as the ITOps and FinOps lead. Until September 2024, he was a principal Customer Success

Manager at IBM. Debasish's work successfully established several advanced IBM technologies in the field and has generated several hundred million USD in revenue for IBM.

Debasish's present areas of interest are superior observability to reduce the mean time to recovery, automatic application resource management, FinOps, and the use of AI to enhance these technologies. Debasish obtained his Ph.D. in combinator-based functional programming languages. He is the father of two brilliant software engineer daughters, Cheenar and Neehar, who are his pride and joy.

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Finally, I would like to acknowledge the entire **Apress staff** who contributed to the creation of this book. Your professionalism, support, and collaborative efforts were essential in bringing this project to fruition. Thank you for your hard work and commitment to excellence.

This book is a result of the collective efforts of all these remarkable individuals, and I am truly thankful for their contributions.

CHAPTER 1

Introduction to Generative AI and Large Language Models (LLMs)

In the constantly evolving realm of artificial intelligence (AI) and machine learning (ML), few advancements have been as revolutionary in recent years as the ascent of Generative AI and Large Language Models (LLMs). These cutting-edge models have redefined our comprehension of AI systems, unleashing a wave of unprecedented opportunities and applications spanning diverse fields. In this chapter, we will embark on an enlightening exploration of the underpinnings, evolution, and profound impact of Generative AI and Large Language Models. Moreover, we will examine how Spring AI seamlessly integrates into this dynamic landscape, shaping the future of AI-driven solutions.